Last week:

- Planar graphs
- Planarity and chromatic numbers

This week:

- Ramsey numbers

Motivation:

- There is a party with N random guests.
- Want to ensure at least 5 know each other **or** that at least 5 are strangers.

What should N be? What is the smallest possible value for N?

Build an N-vertex graph (vertices = guests)

Red edge if two guests know e/o

Blue edge otherwise.

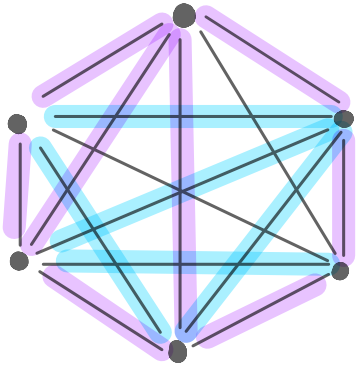
Thus we have K_N with edges colored by red / blue

Want: either blue K_5 or red K_5.

Definition. *Fix an integer* $N \geqslant 2$ *and a 2-edge coloring* $f\colon E(K_N) \to \{red, blue\}$ *for the clique* K_N. *For a given graph* G *we say that:*

- *The coloring* ***contains a red*** G *if* K_N *contains a copy of* G *whose edges are red.*
- *The coloring* ***contains a blue*** G *if* K_N *contains a copy of* G *whose edges are blue.*
- *The coloring* ***contains a monochromatic*** G *if it contains either a red or a blue* G.

Example.



Have a red G-cycle
red K_3
red C_4

Blue K_3
Blue P_4

Note: The motivating problem translates to: Find N such that any red, blue coloring of $E(K_N)$ contains either a blue K_5 or a red K_5

Ramsey theory: what monochromatic structures **must** appear in **any** red-blue coloring of $E(K_N)$.

Definition. *For positive integers m, n, the **Ramsey number** $R(m,n)$ is the smallest integer N such that every red,blue-coloring of $E(K_N)$ contains either a red copy of K_m or a blue copy of K_n.*

Note 1: A red,blue-coloring of $E(K_N)$ corresponds to an N-vertex graph G and its complement $\overline{G}$.

Red-blue coloring of $E(K_N)$ $\rightsquigarrow$ Take G using red edges, $\overline{G}$ using blue

Given G on N-vertices $\longrightarrow$ Define $f: E(K_N) \longrightarrow \{r, b\}$, color red for edges in G blue for edges in $\overline{G}$.

Note 2: $R(m,n)$ is the smallest integer N such that for every N-vertex graph G, either G contains a copy of K_m or $\overline{G}$ contains a copy of K_n.

N-vertex graph G $\longleftrightarrow$ corresponds To a coloring

Note 3: $R(m,n)$ is the smallest integer N such that for every N-vertex graph G, either $\omega(G) \geqslant m$ or $\alpha(G) \geqslant n$.

$\omega(G) \geqslant m$ means, G contains K_m

$\alpha(G) = \omega(\overline{G}) \geqslant n$ means: $\overline{G}$ contains K_n.

Notes:

(1) A priori, it is not clear that $R(m,n)$ exists for all values of m, n.

(2) If $N = R(m,n)$ exists and M is any integer with $M \geqslant N$, then also every red,blue-coloring of $E(K_M)$ contains either a red copy of K_m or a blue copy of K_n.

$f: E(K_M) \longrightarrow \{r,b\}$ restricts to $f: E(K_N) \longrightarrow \{r,b\}$

(3) $R(m,n) = R(n,m)$.

Flip colors!

(4) $R(1,n) = 1$ for every n.

Any r,b coloring of $E(K_1)$ contains a red K_1.

(5) $R(2,n) = n$ for every n.

Take any $f: E(K_n) \longrightarrow \{r,b\}$. Then $\rightarrow$ There is a red edge — red K_2
$\rightarrow$ All edges are blue — blue K_n

Thus $R(2,n) \leqslant n$. If we define $c: E(K_{n-1}) \longrightarrow \{r,b\}$
$c(e) =$ blue, no red K_2, no blue K_n.

(6) If $m_1 \leqslant m_2$ and $n_1 \leqslant n_2$, then $R(m_1,n_1) \leqslant R(m_2,n_2)$.

Keep in mind during proofs: Given an integer N,

(1) In order to prove that $R(m,n) \geqslant N$, we must **construct a** red,blue-coloring of $E(K_{N-1})$ which has neither a red copy of K_m nor a blue copy of K_n.

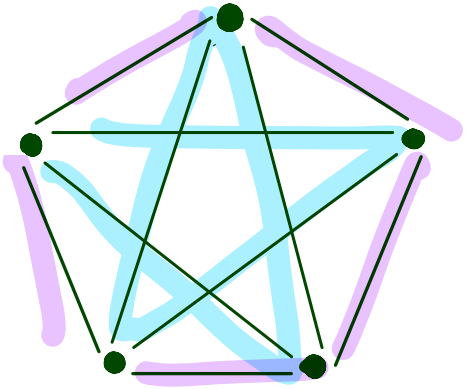
(2) In order to prove that $R(m,n) \leqslant N$, we must **show that every** red,blue- coloring of $E(K_N)$ contains either a red copy of K_m or a blue copy of K_n.

Theorem. $R(3,3) = 6$.

$R(3,3) \geqslant 6$ | construct $f : E(K_5) \longrightarrow \{r,b\}$ with no red K_3, no blue K_3

Red graph $\cong C_5$, no red $\triangle$

Blue graph $\cong C_5$, no blue $\triangle$.



$R(3,3) \le 6$ Any red-blue coloring of K_6 has a monochromatic Δ.

Fix $f: E(K_6) \to \{r, b\}$. Fix $v \in V(K_6)$, and consider

$R = \{x \in V(K_6) : f(vx) = r\}$

$B = \{x \in V(K_6) : f(vx) = b\}$

Thus $N(v) = R \cup B$. Since $5 = |N(v)|$, either $|R| \ge 3$ or $|B| \ge 3$

Wlog, $|R| \ge 3$

- If some $x \neq y$ R have $f(xy) = r$, then $\{v, x, y\}$ is red Δ. ✓
- If all $x \neq y \in R$ have $f(xy) = b$, then any 3-vertices in R ✓ form a blue Δ.

Case $|B| \ge 3$: same argument, flip colors

□

Theorem. *For integers* $m, n \geqslant 2$, *if* $R(m-1,n)$ *and* $R(m,n-1)$ *both exist, then* $R(m,n)$ *also exists and*

$$R(m,n) \leqslant R(m-1,n) + R(m,n-1).$$

Proof: Use $N := R(m-1,n) + R(m,n-1)$.

Take any $f: E(K_N) \longrightarrow \{r,b\}$.

WTS: either f contains a red K_m or a blue K_n.

Fix $v \in V(K_N)$, define

$R = \{x : f(vx) = r\}$, $B = \{x : f(vx) = b\}$, Thus

$N(v) = R \sqcup B$, $\Rightarrow$ $N-1 = |N(v)| = |R| + |B|$

Hence either $|R| \geqslant R(m-1,n)$ or $|B| \geqslant R(m,n-1)$

If $|R| \geqslant R(m-1,n)$. Restrict f to the red graph (induced $K_6[R] \cong K_{|R|}$). Since $|R| \geqslant R(m-1,n)$, this restricted coloring

for $K_6[R]$ contains either a red K_{m-1} or a blue K_n

- If it contains a blue K_n ✓
- If it contains a red K_{m-1}, say using vertices R'

Then $R' \cup \{v\}$ is a red K_m in K_N.

If $|B| \geq R(m, n-1)$, same argument, flip colors. ∎

Corollary. *For integers m, n, the Ramsey number $R(m,n)$ exists and*

$$R(m,n) \leqslant \binom{m+n-2}{m-1} = \binom{m+n-2}{n-1}.$$

Proof: Use Pascal's: $\binom{t}{k} = \binom{t-1}{k-1} + \binom{t-1}{k}$

Induction on n, m.

Base case: $n=1$ or $m=1$, $R(m,n) = 1 \leq \binom{m+n-2}{0} = 1$

Assume $m, n \geqslant 2$. By IH, $R(m-1,n)$ and $R(m,n-1)$ exist.

$$R(m,n) \underset{\text{Previous Thm}}{\leq} R(m-1,n) + R(m,n-1) \leq \binom{m+n-3}{m-2} + \binom{m+n-3}{m-1} \underset{\text{Pascal's}}{=} \binom{m+n-2}{n-1}$$

Corollary. *For any positive integer* n*, we have*

$$R(n,n) \leqslant \binom{2n-2}{n-1} \leqslant 2^{2n-2} \leqslant 4^n.$$

"Proof": $R(n,n) \leq \binom{2n-2}{n-1}$ follows from Previous. ☐

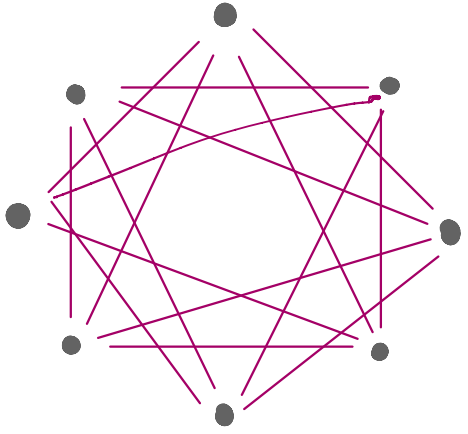
$R(n,n) \leq 4^n \quad \forall n$

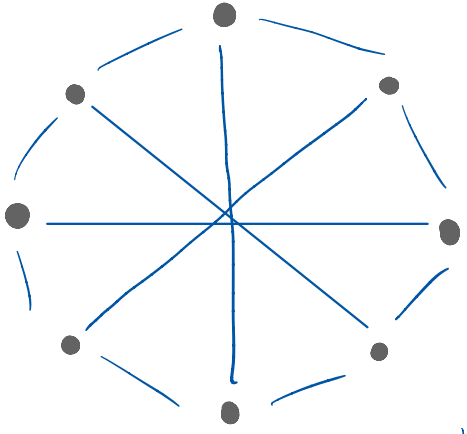
No bound of the form $R(n,n) \leq (4-\varepsilon)^n$ $\forall n$ is known

$R(4,3) \leq R(3,3) + R(4,2) = 6 + 4 = 10$

Theorem. $R(4,3) = 9$.

Proof: $(R(4,3) \geq 9$ construct a $f : E(K_8) \longrightarrow \{r,b\}$ such that there is no red K_4, no blue K_3





$H_{3,8}$

If $V(K_8) = \{v_0, \ldots, v_7\}$ we color $v_i v_j$ blue if $i - j \in \{\pm 1, 4\} \bmod 7$

$v_i v_j$ red otherwise

$R(4,3) \leq 9$ | Take any $f : E(K_9) \longrightarrow \{r, b\}$

WTS: f contains either red K_4 or a blue K_3

Fix $v \in V(K_9)$. Let $N(v) = R_v \cup B_v$ as before.

using the previous argument,

if $|R| \geq \underbrace{R(3,3)}_{6}$ or $|B| \geq \underbrace{R(4,2)}_{4}$, we know how to win.

Can assume $|R| \leq 5$ and $|B| \leq 3$ for any v.

BUT $8 = |N(v)| = |R_v| + |B_v| \Rightarrow |R_v| = 5,\ |B_v| = 3$.

Thus $G_r = (V(K_9),$ red edges$)$ is 5-regular

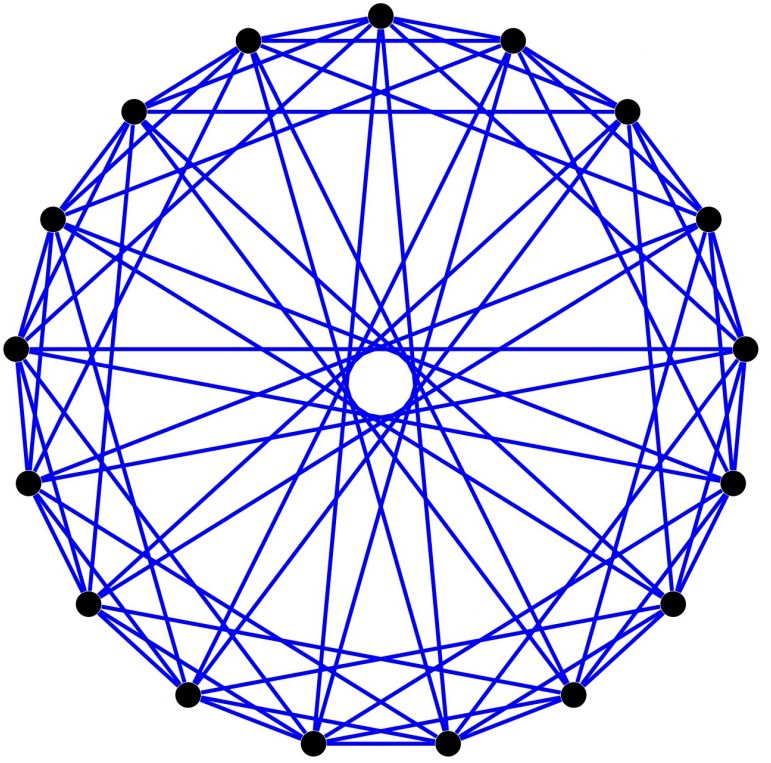
G_r has an odd number (9) of odd-vertices

contradicts Handshaking. ▨

Theorem. $R(4,4) = 18$.

K_{17}

"Paley graphs"



Courtesy Chris Cox

$i-j \in \{\pm 1, \pm 2, \pm 4\}$

Best available bounds:

$$43 \leqslant R(5,5) \leqslant 48, \qquad 102 \leqslant R(6,6) \leqslant 165.$$